EXP NO:20 EXPERIMENT TO UNDERSTAND K-MEANS CLUSTERING ALGORITHM FOR A GIVEN DATA SET

CODE :

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
df=pd.read_csv('Mall_Customers.csv')
df.info()
df.head()
sns.pairplot(df)
features=df.iloc[:,[3,4]].values
from sklearn.cluster import KMeans
model=KMeans(n_clusters=5)
model.fit(features)
KMeans(n_clusters=5)
Final=df.iloc[:,[3,4]]
Final['label']=model.predict(features)
Final.head()
sns.set_style("whitegrid")
sns.FacetGrid(Final,hue="label",height=8) \
.map(plt.scatter,"Annual Income (k$)", "Spending Score (1-100)") \
```

```
.add_legend();
plt.show()
features_el=df.iloc[:,[2,3,4]].values
from sklearn.cluster import KMeans
wcss=[]
for i in range(1,10):
model=KMeans(n_clusters=i)
model.fit(features_el)
wcss.append(model.inertia_)
plt.plot(range(1,10),wcss)
```

OUTPUT :

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 600 entries, 0 to 599
Data columns (total 5 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   CustomerID              600 non-null    int64
 1   Gender                  600 non-null    object
 2   Age                     600 non-null    int64
 3   Annual Income (k$)      600 non-null    int64
 4   Spending Score (1-100)  600 non-null    int64
dtypes: int64(4), object(1)
```

memory usage: 23.6+ KB

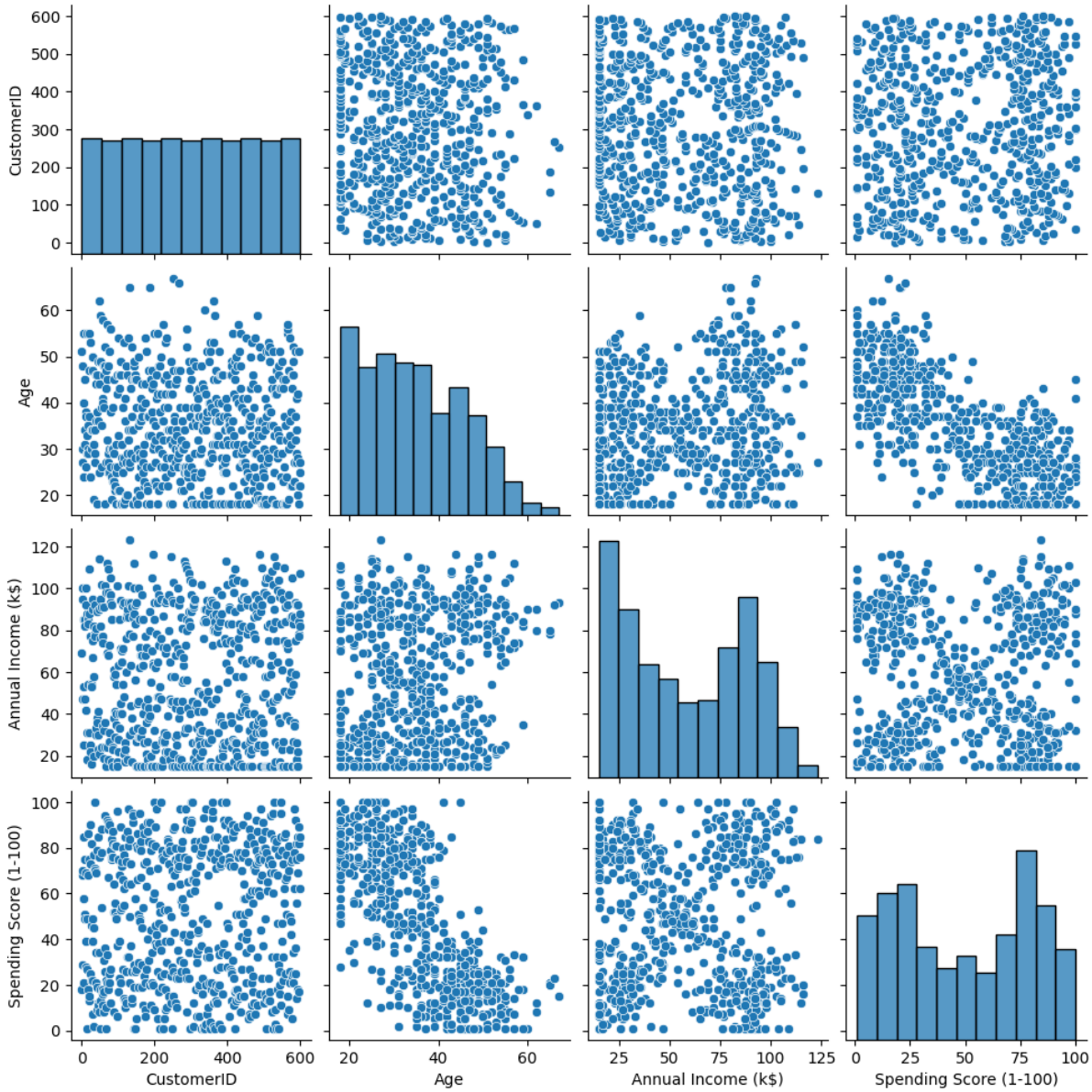
/tmp/ipykernel_6488/228583201.py:16: SettingWithCopyWarning:

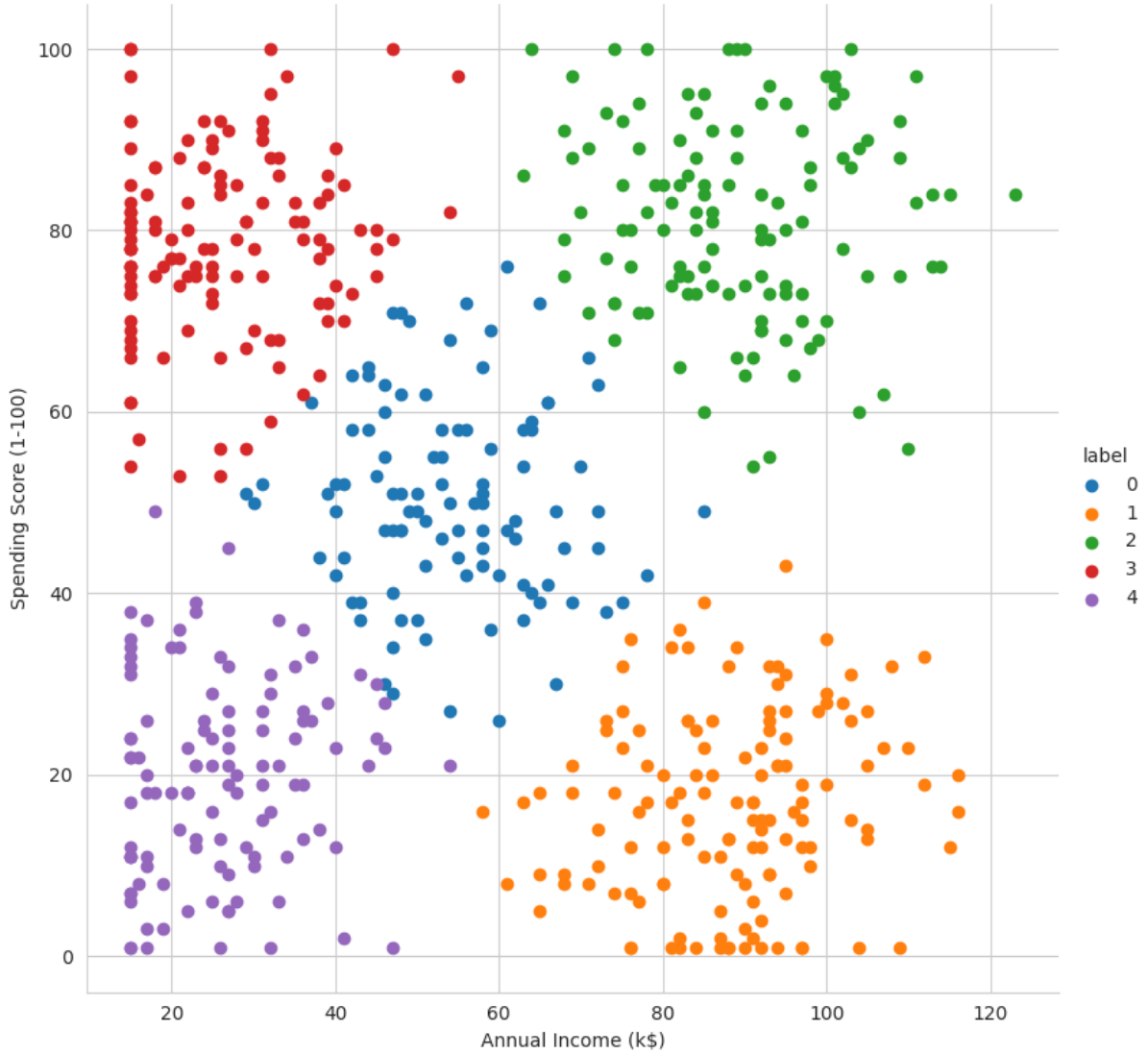
A value is trying to be set on a copy of a slice from a DataFrame.

Try using .loc[row_indexer,col_indexer] = value instead

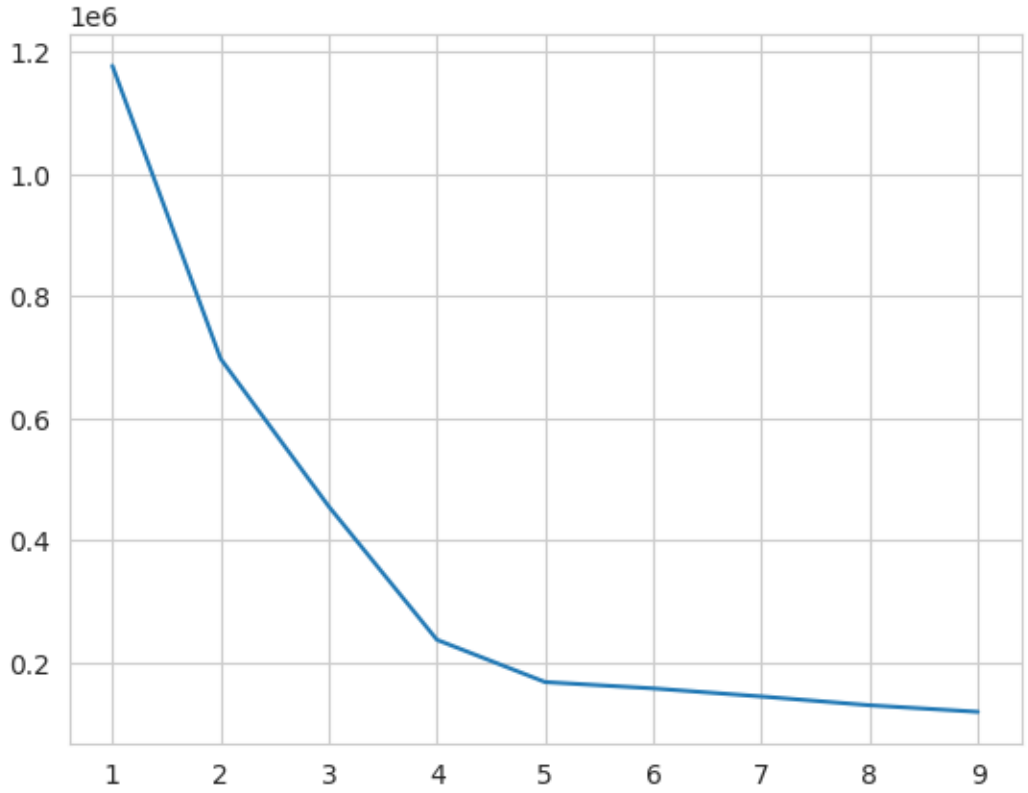
See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
  Final['label']=model.predict(features)
```


CustomerID
Age
Annual Income (k$)
Spending Score (1-100)
0
200
400
600
20
40
60
25
50
75
100
125
0
25
50
75
100



[<matplotlib.lines.Line2D at 0x7e34dc269a90>]


1e6
1.2
1.0
0.8
0.6
0.4
0.2
1
2
3
4
5
6
7
8
9